UG (NEP) 5th SEMESTER
COMPUTER APPLICATIONS – DISCRETE MATHEMATICS
Course Code: CAP522J3

Session: December–January 2024–25 No. of Questions: 09
Time Allowed: 2:00 Hours Maximum Marks: 100 Minimum Marks: 36

NOTE: Attempt all questions from Sections A and B and only two questions from Section C.

SECTION A: Short Answer Type Questions (8 × 2 = 16 Marks)

Answer all questions in a maximum of 20 words each.

1. Find the negation of statement: “For all integers n, either n is even or n is odd.”
2. In how many ways can we select a team of 4 students from a given choice of 15?
3. Let G be a simple graph with 40 edges and its complement G′ has 38 edges. Find the number of vertices in graph G.
4. Is it possible for a planar graph to have 6 vertices, 10 edges and 5 faces? Explain.
5. For the function $y = 2x + 1$, find the range when domain = $\{-3, -2, -1, 0, 1, 2, 3\}$.
6. Let U be the set of real numbers, $A = \{x \mid x \text{ is a solution of } x^2 - 4 = 0\}$ and $B = \{-1, 4\}$. Compute $A \cup B$ and $A \cap B$.
7. With the help of a truth table show that $(p \rightarrow q) \equiv (\neg p \vee q)$.
8. State De Morgan’s laws for quantifiers.

SECTION B: Medium Answer Type Questions (4 × 10 = 40 Marks)

Answer all four questions with internal choice in a maximum of 150 words each.

2. A survey among 100 students shows that, of the three ice-cream flavours vanilla, chocolate and strawberry, 50 students like vanilla, 43 like chocolate, 28 like strawberry. There are 13 students who like vanilla and chocolate, 11 like chocolate and strawberry, 12 like strawberry and vanilla, and 5 like all three. Find the number of students who like each of the following flavours:

   i) Chocolate but not strawberry. ii) Vanilla or chocolate but not strawberry.

OR

Among the 30 students registered for a course in discrete mathematics, 15 students know Java, 12 know C++, and 5 know both languages. Find:

   i) How many students know at least one of Java or C++? ii) How many know only C++? iii) How many know exactly one of the languages Java and C++?

3. Prove or disprove the statement that $(p \wedge q) \wedge (p \rightarrow q)$ is a tautology.

OR

Determine the validity of the following argument:

   If I work hard, then I earn lots of money. If I earn lots of money, then I pay high taxes. Therefore, if I do not work hard, then I do not pay high taxes.

4. In a 12-day period, a small business mailed 195 bills to customers. Using the Pigeonhole Principle show that during some period of three consecutive days at least 49 bills were mailed.

OR

Consider all 3-digit numbers 000 to 999. In how many of these numbers are all the digits different?

5. Prove that the number of odd-degree vertices in a graph is an even integer.

OR

Show that K4 and K2,2 are planar graphs.

SECTION C: Long Answer Type Questions (2 × 22 = 44 Marks)

Answer only two questions in a maximum of 400 words each.

6. Show that $\neg(p \rightarrow q)$ is equivalent to $p \wedge \neg q$.
7. Prove that “A simple graph is connected if and only if it has a spanning tree.”
8. Let P(x), Q(x), and R(x) be the statements “x is a professor,” “x is ignorant,” and “x is vain,” respectively. Express each statement using quantifiers, logical connectives, and P(x), Q(x), and R(x), where the domain consists of all people:

   i) No professors are ignorant. ii) All ignorant people are vain. iii) No professors are vain. iv) Does (iii) follow from (i) and (ii)?

9. Prove by contradiction that there is no greatest even integer.